

# AADE Filter Design V4.5: A Design and Analysis Tool for Passive Filters

Here is freeware for the design and analysis of passive filters in both the time and frequency domains. It is easy to use, supports different types of filters, runs on Windows, and has several powerful transformations along with an extensive help file. The AADE Filter Design program was originally commercial but is now free

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**A**ADE Filter Design is simple, basic software that aids circuit designers and students to design filters and analyse several of its parameters in both the time and frequency domains. It is a project written exclusively by Neil Heckt of Almost All Digital Electronics (AADE).

This software has built-in templates for about a dozen common and less common crystal filter types—Butterworth, Chebyshev, elliptic, Bessel, Legendre, linear phase, coupled resonator and crystal ladder. One can select the configuration (whether high-pass, low-pass, band-pass or band-reject) and enter the specifications with which the filter is to be designed. The user can analyse and alter any filter design by entering a schematic and parts list. It is possible to insert a new filter in front of an existing filter or append a new filter at the end. This feature allows the user to create a wide band-pass filter with low-pass/high-pass filter combinations.

There are several other attributes to this program. You can specify the unloaded quality factor (Qu) for inductors and, hence, get response plots corresponding to how these components behave in the real world. The software designs air-wound and toroid coils, and n-type and T-type attenuators. The transformation of design impedance

and/or design frequency is possible. The software enables the setting of components to the nearest specified tolerance and nearest standard value. There are three in-built options available for statistical analysis: Monte Carlo, worst case and sensitivity analysis.

methods. The software has several powerful transformations that enable the user to modify component values, thus making the filter easier to build and align. All files are saved in the \*.sch format and you can print the design for analysis.



Fig. 1: Main window of AADE Filter Design

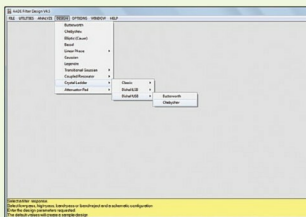


Fig. 2: Drop down menu showing built-in filter response templates

## Filter types

This tool can be used to design and analyse a wide variety of passive filters.

It allows the design of linear filters like Butterworth, Chebyshev, elliptic (Cauer), Bessel, Legendre and linear phase. These filters can be of the high-pass, low-pass, band-pass or band-reject type.

Coupled resonator band-pass filters and crystal ladder band-pass filters are also supported by AADE Filter Design.

## Signal processing

The software offers both time domain and frequency domain analysis.

The time domain